CSMFAB000000000038 Aegis Embark_ Haptic Insoles Fabrication

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** `CSMFAB000000000038 Aegis Embark_ Haptic Insoles Fabrication V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
Piezoelectric material	V1.0: PZT-5H; d33=289 pC/N; Tc=193°C; lead-containing	V2.0: KNbO3-BaTiO3; d33=285 pC/N; Tc=310°C; lead-free	RoHS/REACH compliance — PZT restricted; Tc +117°C prevents thermal depolarization	TDK 2025 lead-free catalog; J. Am. Ceram. Soc. 108 (2025)
Post-event navigation	V1.0: GPS assumed operational	V2.0: LoRa mesh 915 MHz, 256-node, 2-5 km/hop; MRAM offline maps	GPS vulnerable to solar energetic particle damage in Carrington scenario	ITU-R WP 4C 2024; LoRa Alliance v1.1

Parameter	V1.0	V2.0	Reason	Primary Source
ZrB2-SiC sintering temperature	V1.0: SPS at 1900–2100°C, 30–50 MPa, 20 min	V2.0: Flash Sintering at 1800°C, 50 MPa, 5 min; >98% TD achieved	40% energy reduction; lower T preserves FSS layer integrity; equivalent density	Johnson et al., Acta Materialia 278 (2024) 120223

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
Piezoelectric material	Lead restricted under RoHS Annex II; Tc 117°C lower = premature depolarization risk	HIGH
Post-event navigation	GPS invalid assumption in Carrington scenario; navigation fails in primary use case	HIGH
ZrB2-SiC sintering temperature	Excess 400°C degrades embedded FSS; higher energy cost vs V2.0 process	MEDIUM

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000038 Aegis Embark_ Haptic Insoles Fabrication V1.0 archived; V2.0 is the active specification as of June 2026.